

Functional Programming

1. Greeting with Lambda

Task:

Create a functional interface `Greeting` with a method `void greet(String name)`.

Instructions:

- Implement it using a lambda to print: `"Hello, <name>!"`
 - Call it with multiple names in the `main()` method.
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2. Math Operations with Lambdas

Task:

Create a functional interface `Operation` with `int apply(int a, int b)`.

Instructions:

- Implement lambdas for:
 - Addition
 - Subtraction
 - Multiplication
 - Division
- Write a method `calculate(Operation op, int a, int b)` to use it.

3. Simple String Processor

Task:

Define a functional interface `StringProcessor` with `String process(String input)`.

Instructions:

- Write 3 lambda implementations:
 - Convert to uppercase
 - Reverse a string
 - Replace spaces with dashes
- Call and print output for each.

4. Lambda for Array Search

Task:

Create a functional interface `Checker` with method `boolean check(int x)`.

Instructions:

- Implement a method `countMatches(int[] arr, Checker checker)`
- Pass lambdas to count:
 - Even numbers
 - Numbers greater than 100
 - Negative numbers

5. Simple Conditional Message

Task:

Create a functional interface `Condition` with `boolean test(int value)`.

Instructions:

- Use a lambda to print "Pass" or "Fail" based on whether value > 50.
- Test this in a loop from 1 to 100.

6. Character Filter in String

Task:

Define a functional interface `CharFilter` with `boolean accept(char c)`.

Instructions:

- Implement a method `printAcceptedChars(String s, CharFilter filter)`
- Use lambdas to:
 - Print only vowels
 - Print only uppercase letters
 - Print only digits

7. Lambda with Static Method

Task:

Use a functional interface `Printer` with `void print()`.

Instructions:

- Create a static method `static void printStars()` that prints `*****`
- Assign it to a lambda or method reference and call `print()`

8. Temperature Conversion

Task:

Define a functional interface `Converter` with `double convert(double value)`.

Instructions:

- Use lambdas to convert:
 - Celsius to Fahrenheit
 - Fahrenheit to Celsius
- Call each with example values and print results.

9. Length Comparison

Task:

Create a functional interface `CompareLength` with `boolean isLonger(String s1, String s2)`.

Instructions:

- Implement a lambda to compare string lengths.
- Call it with various string pairs to find which is longer.

10. Simple Timer with Lambda

Task:

Create a functional interface `TimerAction` with `void run()`.

Instructions:

- Write a method `runWithDelay(TimerAction action, int delayMs)` that:
 - Waits for `delayMs` milliseconds
 - Then runs the `action.run()` lambda
- Use this to print messages after short delays.